

DISCOVERY OF ETI: TERRESTRIAL AND EXTRATERRESTRIAL LEGAL IMPLICATIONS

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Abstract—The Legal Situation on Earth: The following international legal regulations seem to apply to the search for and the eventual detection of ETI: a) The "Space Treaty" of Oct. 10, 1967; b) The Liability Convention of Oct. 9, 1973; c) The Moon Agreement of Dec. 5, 1979; d) The International Telecommunication Convention.

Legal Relations with Extraterrestrials: We may expect the following characteristics of ETI: 1. life in the sense of influencing the environment by selection from more than one possibility; 2. intelligence in the sense of self-realization of free will; 3. existence in three dimensional space and a will to live.

With this we can expect that each race in the universe may have the following interests regarding its own race: a) to preserve and continue its own life; b) to protect this life from damage and intrusion; c) possibly to expand the realms of its living space.

Therefore, if we decide to "answer" ETI, we may want to transmit such legal-philosophical principles: 1. the principle of nonviolation; 2. the principle of equality; 3. the principle to recognize the will to live and the living space of any intelligent race.

1. INTRODUCTION

For more than 25 years now, the search for extraterrestrial intelligence has been going on. The search has mostly been conducted by trying to find alien electromagnetic emissions. On the other hand, at least four spaceprobes have been provided with a kind of official "letter" to be possibly found at some future date by ETI[1,2]. Thus, mankind has tried to listen and to speak to ETI. Up to now, nobody has questioned the legality of such endeavors. However, there do exist already international legal rules applicable to SETI[3].

These rules are, of course, of terrestrial origin, and norms of *lex lata*[4]. They will have to be examined in detail.

Furthermore, we should investigate possible extraterrestrial legal implications. This means that we should investigate whether ETI might be legal subjects, such as we humans and our nations are, and which kind of sociological and legal situations might arise if we answer this question.

2. THE LEGAL SITUATION ON EARTH

About two decades ago, nearly all states within the framework of the United Nations decided to legally regulate their activities regarding Outer Space. There are several treaties and agreements which are to be considered concerning our topic. Doing this we will have once more to remember that they are terrestrial international laws, therefore applicable to Earth nations and peoples and not

to any possible ETI. A detailed investigation results in the following observations:

1. There is in force the "Treaty on Principles Governing the Activities of States in the Exploration and use of Outer Space, including the Moon and Other Celestial Bodies" of Oct. 10, 1967. This Treaty - and there is not the slightest doubt about it among statesmen and lawyers - comprises binding International Law, to be adhered to by practically all nations and their organizations. This Treaty then, bears the following provisions:

"The explorations and use of Outer Space....shall be carried out for the benefit and in the interest of all Countries....and shall be the province of all mankind[5].

"There shall be freedom of scientific investigation in Outer Space, including the Moon and other Celestial Bodies, and States shall facilitate and encourage international cooperation in such investigation[6].

"States Parties to the treaty shall regard astronauts as envoys of mankind in Outer Space....[7].

"States Parties to the Treaty shall immediately inform the other State Parties of the Treaty or the Secretary-General of the United Nations of any phenomena they discover in Outer Space, including the Moon and other Celestial Bodies, which could constitute a danger to the life or health of astronauts [8].

"States Parties to the Treaty shall bear international responsibility for national activities in Outer Space, including the Moon and other Celestial bodies,

whether such activities are carried out by governmental agencies or by non-governmental entities, and for assuring the national activities are carried out in conformity with the provisions set forth in the present Treaty. The activities of non-governmental entities in Outer Space, including the Moon and other Celestial Bodies, shall require authorization and continuing supervision by the appropriate State Party on the Treaty[9].

"If a State Party to the Treaty has reason to believe that an activity or experiment planned by it or its nationals in Outer Space, including the Moon and other Celestial bodies, would cause potentially harmful interference with activities to other States Parties in the peaceful exploration and use of Outer Space, including the Moon and other Celestial bodies, it shall undertake appropriate international consultations before proceeding with any activity or experiment[10].

"States Parties to the Treaty conducting activities in Outer Space, including the Moon and other Celestial Bodies, agree to inform the Secretary General of the United Nations as well as the Public and the international scientific community, to the greatest extent feasible and practicable, of the nature, conduct, locations and results of such activities. On receiving the said information, the Secretary General of the United Nations should be prepared to disseminate it immediately and effectively[11]."

This means, in short: nations are responsible to other nations regarding their activities and the activities of their organizations and their inhabitants. And if such activities might be harmful, consultations with other nations are obligatory.

The notion of "benefit to all mankind" demonstrates a mutual obligation of all terrestrial States. This obligation may acquire a new dimension if extraterrestrial entities actually are discovered.

And when speaking of astronauts as "envoys of mankind" there immediately rises the question "envoys to whom?" This notion also indicates a special coherence between the members of "mankind" toward Outer Space and its (possible) inhabitants.

2. The "Convention on International Liability for damage caused by Space Objects" of

Oct. 9, 1973 although it mainly deals with other topics, contains the following provisions:

"A state, which suffers damage or whose natural or juridical persons suffer damage, may present to a launching state a claim for compensation for such damage[12].

"If the damage caused by a space object presents a large scale danger to human life or seriously interferes with the living conditions of the population or the functioning of vital centers, the States Parties, and in particular the launching state, shall examine the possibility of rendering appropriate and rapid assistance to the state which has suffered the damage, when it so requests[13]."

This means that nations have to compensate for damage which might be caused by Outer Space activities of their organizations and inhabitants. We should, therefore, bear in mind that a contact with ETI might constitute some possibility of damage. And we might wish to be careful on this account.

3. Finally, the "Agreement Governing the Activities of States on the Moon and Other Celestial Bodies" of Dec. 5, 1979 (although not yet ratified by either the USA or the USSR, but in force since 1984) bears the following provisions:

"The exploration and use of the Moon shall be province of all mankind and shall be carried out for the benefit and the interests of all countries, irrespective of their degree of economic or scientific development. Due regard shall be paid to the interests of present and future generations as well as to the need to promote higher standards of living conditions of economic and social progress and development in accordance with the Charter of the United Nations[14].

"In carrying out activities under this Agreement, States Parties shall promptly inform the Secretary General, as well as the public and the international scientific community, of any phenomena they discover in Outer Space, including the Moon, which should endanger human life or health, as well as of any indication of organic life[15].

"States Parties shall also take measures to prevent harmfully affecting the environment of the Earth through the introduction of extraterrestrial matter or

otherwise[16]. "States Parties shall report to other States Parties and to the Secretary General concerning areas of the Moon having special scientific interest in order that, without prejudice to the rights of other States Parties, consideration may be given to the designation of such areas as international scientific preserves for which special protective arrangements are to be agreed in consultation with the competent organs of the United Nations[17]."

Here, finally, the lawmakers of this treaty clearly foresaw that extraterrestrial life might exist and might be dangerous to human life, and on the other hand that phenomena of extraterrestrial life might be worth preserving.

The authors of International Law, therefore, proved to be careful; when conducting SETI, nations and their citizens ought to exercise some care, too.

4. Furthermore, the International Telecommunication Convention provides that the International Telecommunication Union (ITU) has among others the purpose, "to maintain and extend international cooperation for the improvement and rational use of telecommunications of all kind and to promote the adoption of measures for ensuring the safety of life through the cooperation of telecommunication services[18]."

5. One of the permanent organs of the ITU is the International Radio Consultative Committee (CCIR). Among its duties is "the peaceful uses of Outer Space regarding Space Research." It is expressly stated that "radio astronomy" belongs to this kind of activity, including the "reception of extraterrestrial radiation, caused by physical phenomena[19]."

Special portions of the electromagnetic spectrum are considered of interest for SETI and terrestrial activities of other parties should not impede this search.

3. LEGAL RELATIONS WITH EXTRATERRESTRIALS?

We can only speculate regarding the physical and psychical (intellectual) characteristics of alien life forms. However, if any contact at all is possible, some basic characteristics are necessarily to be expected. Those characteristics can be deduced from the basic qualities of life and of intelligence themselves.

Life is a phenomenon of physical nature. The living organism copes with and masters its

environment. This environment is the physical inorganic universe, which is subject to entropy.

An organism will be considered as living if it is able to select between several possibilities. It will select that possibility which is least harmful. Generally speaking we may say that a living organism selects the alternative that minimizes damage.

By way of mutation life shows a trend towards the evolution of organisms which are still more able to deal with the environment. Finally, life reaches the highest possible level of evolution, namely the level of intelligence.

Intelligence, or reason, might be defined as the capacity for logical thought and for recognizing the capacity of choosing between two or more possibilities and of self-consciousness. Like all living beings, the intelligent ones - but consciously so - will choose the alternatives which seem to be least harmful to them.

The greatest harm which can befall a being is its extinction, its death. Therefore, the selection of possibilities will practically always go into the direction of preserving the organism's existence. This brings about the most basic characteristics of life, namely, the will to live.

Intelligent life thus recognizes the freedom of will and recognizes furthermore its unalterable will to live and to protect its life. Thus, an intelligent living organism will be able to realize that harmful acts can be committed.

At the same time it will be able to realize that harmful acts can be suffered as well. Therefore, it would consider harmful acts committed against it as undesirable or, in other words, as "evil." And it will consider acts which protect it from damage as desirable or "good."

Of course, the "will to live" might not be restricted to one individual alone. It might be extended to one's offspring, to one's mates, friends, parents (if any), to the whole group or the whole race.

Furthermore, by definition, the human race will only be able to contact other life forms which are in some way detectable. And they will be detectable only if they exist or at least partly operate within our three dimensional universe.

These considerations regarding the basic notions of life, intelligence, the will to live and the space in which to live bring about basic characteristics for all sentient beings whom we may encounter. Those characteristics might thus be enumerated as follows[20]:

1. life, in the sense of influencing the environment by selection from more than one possibility, that is, by fighting entropy;

2. intelligence, in the sense of self-realization of free will, and realization of the basic ideas of good and evil;
3. detectability by humans;
4. three-dimensionality, or at least existence or activity within three-dimensional space;
5. some, if only rudimentary, will to live.

Within these characteristics virtually anything is possible; virtually everything must be expected.

These considerations regarding the basic notions of life, intelligence, the will to live and the space in which to live bring about basic characteristics for all (living or extinct) sentient beings of which we might find traces.

We should not be anthropocentric. And in a similar sense, aliens gifted with intelligence will realize that they should not center their deliberations on all characteristics of their own life forms.

However, all intelligent races that might come into contact with each other should probably expect the characteristics quoted above regarding all possible intelligent races in the universe.

With this, all intelligent life forms will have at least some basic interests, which we might enumerate as follows:

1. to preserve and continue its own life and/or the life of its own race;
2. to protect this life from damage and intrusion;
3. possibly to expand the realms of its own life and/or the life of its own race.

4. PEACEFUL CONTACT WITH ETI

Up to now, we only know one kind of life, namely that on Earth. And we know only one kind of intelligence, namely the human one.

Both terrestrial life and intelligence show distinct qualities of the capability of "fighting for survival". The whole evolutionary process of life, creating higher and higher organisms and, finally, organisms equipped with intelligence is known to have been a result of the selection process of the "survival of the fittest". It is therefore impossible to exclude the notion that extraterrestrial intelligent life might be a product of such an evolutionary process as well. With this, the notion of "fighting for survival" the notion of competition and the notion of survival of the fittest may at least be a possible quality or characteristic of such extraterrestrial intelligence.

At the same time, the intelligent extraterrestrial will probably speculate in the same direction. And thus he might realize that mankind could be dangerous to his race.

All this is true, even if we are only able to detect extraterrestrial intelligent signals and if we might possibly answer them. No physical encounter is necessary as long as at least any contact, and with this any competition, is imaginable.

Thus, both the human race and ETI might wish to include statements of their position; we might want to make clear that we do not look for such competition but we look only for contact and mutually respectful relations. Therefore, we might decide to maintain the following positions and even to demand such positions from any other intelligent race. And we might decide to communicate these points of view along with detailed signals. Such communications might be:

"We shall not harm you!"

"We shall not permit you to harm us."

"If harm is done unwittingly, we shall restore your integrity as fully as we can."

"We regard you and ourselves as equals, neither you nor we are superior, neither you nor we are inferior."

"If we promise something, we shall keep our word; if you promise something please do the same."

"We have the will to live; we realize and appreciate that you do as well."

"We need three-dimensional living space; we realize that you have the same need. Therefore we shall not impair your space; please do not impair ours."

"We intend to develop our race, but we recognize that for you the existence of your race has preference over our development. However, we expect you to take the same attitude with regard to us."

"If we can help you in any way, please tell us. It is an ethical principle for us to extend help to you."

The other life form, however far it may have developed, will recognize the sincerity of our communication. It will recognize, furthermore, that we expect the same statements of policy and the same sincerity from them.

Man is a rather aggressive race, but even this race is able to formulate rules of universal validity. If, therefore, the ETI understands our statements but does not transmit the same ideas, then we should assume that it might be more aggressive than we, and we should then even be prepared for an inimical attitude on its part.

A really intelligent extraterrestrial, however, will realize that such attitudes would prove to be counterproductive. Peaceful contact, on the other hand, will most probably be beneficial to all intelligent races in the universe.

REFERENCES

1. ETI - Extraterrestrial Intelligence.
2. Plates engraved with drawings and messages were attached to the Pioneer 10 and 11 Space Probes. (See Sagan, *The Cosmic Connection*, New York, (1973)). The Space Probes Voyager I and Voyager II, both launched in 1977, carried the following message by the UN Secretary General Kurt Waldheim:
 "As the Secretary-General of the United Nations, an Organization of 147 Member States who represent almost all of the human inhabitants of the planet Earth, I send greetings on behalf of the people of our planet. We step out of our solar system into the universe seeking only peace and friendship; to teach if we are called upon; to be taught if we are fortunate. We know full well that our planet and all its inhabitants are but a small part of this immense universe that surrounds us, and it is with humility and hope that we take these steps."
3. *Search for ETI*
4. *Lex lata* - existing law; *Lex ferenda* - law to be created in the future.
5. Art. I (1)
6. Art. I (3)
7. Art. V (1)
8. Art. V (3)
9. Art. VI
10. Art. IX
11. Art. XI
12. Art. VIII (1)
13. Art. XXI
14. Art. IV (1)
15. Art. V (3)
16. Art. VII (1)
17. Art. VII (3)
18. For an overview see UN Document A/AC I05/193, *Space Activities and Resources*. New York, p. 38 (1977).
19. A/AC I05/193, p. 61.
20. E. Fasan, *Relations with Alien Intelligences*. Berlin, p. 37 (1970).